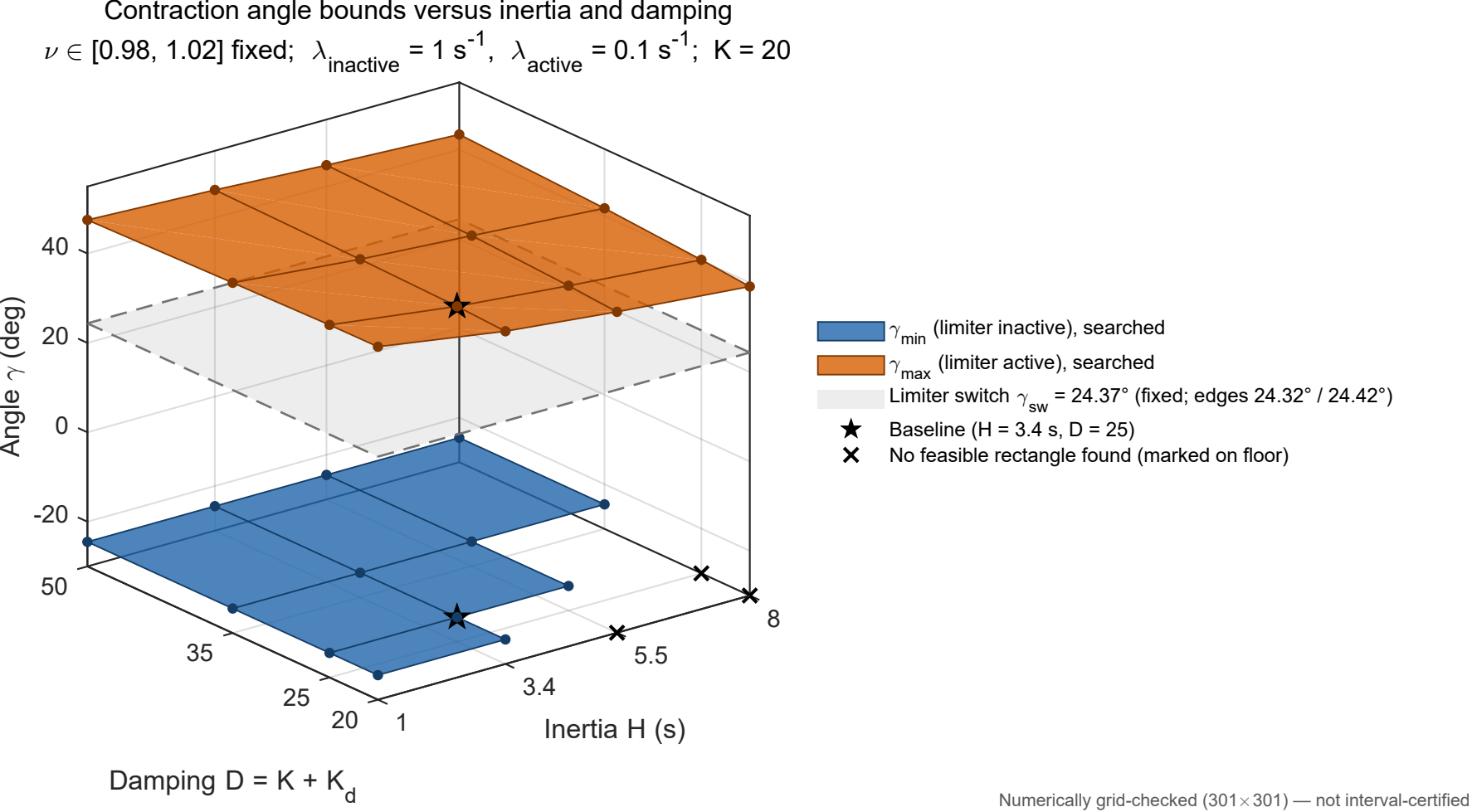
Contraction angle bounds versus inertia and damping
$\nu \in [0.98, 1.02]$ fixed; $\lambda_{inactive} = 1\ s^{-1}$, $\lambda_{active} = 0.1\ s^{-1}$; K = 20
Angle γ (deg)
40
20
0
-20
50
35
25
20
Damping D = K + K_d
1
3.4
5.5
8
Inertia H (s)
γ_{min} (limiter inactive), searched
γ_{max} (limiter active), searched
Limiter switch γ_{sw} = 24.37° (fixed; edges 24.32° / 24.42°)
Baseline (H = 3.4 s, D = 25)
No feasible rectangle found (marked on floor)
Numerically grid-checked (301×301) — not interval-certified